Test 2 review day

CMSE 381

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Dept of Computational Mathematics, Science & Engineering

Fri, March 14, 2025

Announcements

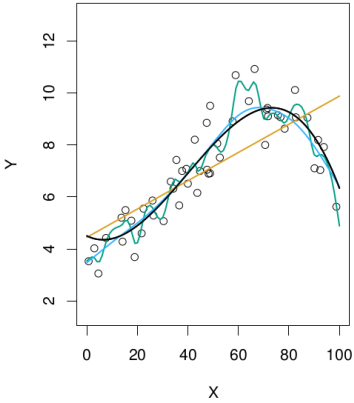
Announcements:

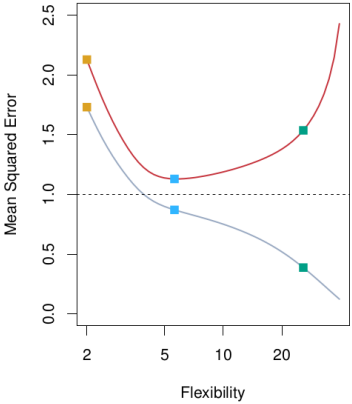
- midterm # 2 on Monday
- HW 5 due Sunday (3/16)

Top questions submitted

- Compare different kinds of validation
- difference between lasso and ridge regression
- go through subset selection example again

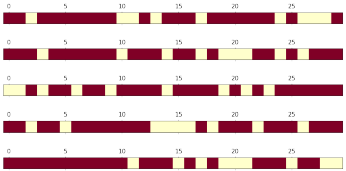
Training error vs test error

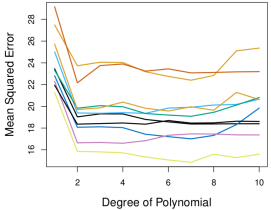




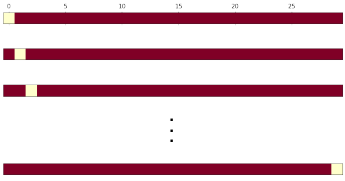
Approximations of Test Error

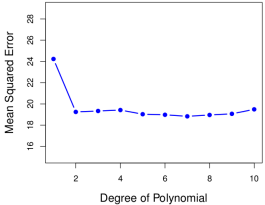
Validation Set



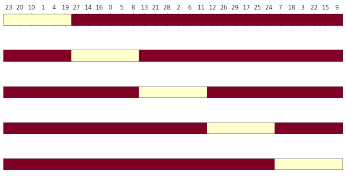


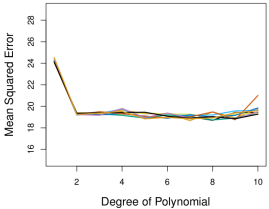
LOOCV



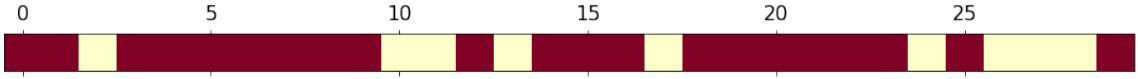


K-fold CV



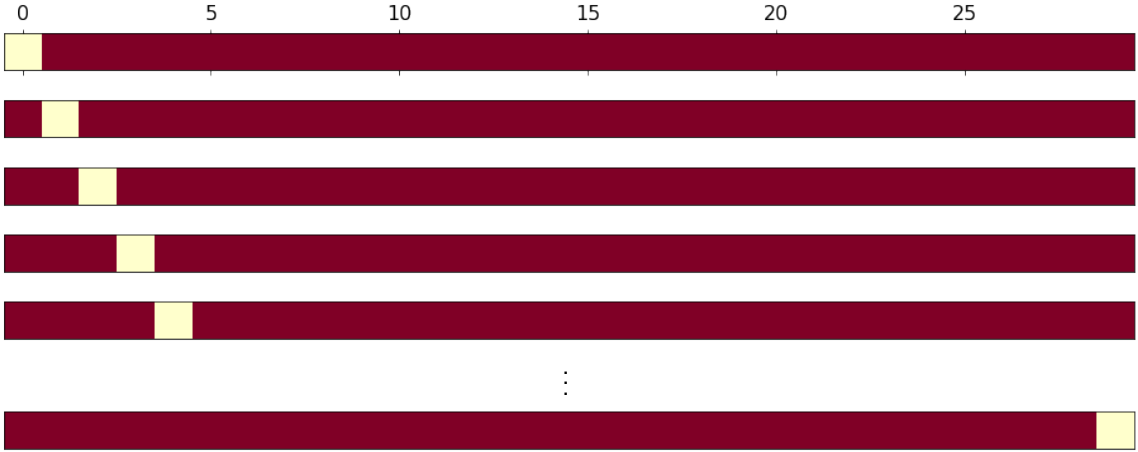


Validation set approach



- Divide randomly into two parts:
  - Training set
  - Validation/Hold-out/Testing set
- Fit model on training set
- Use fitted model to predict response for observations in the test set
- Evaluate quality (e.g. MSE)

LOOCV



LOOCV in mathy words

- Remove (x_1, y_1) for testing.
- Train the model on $n-1$ points: $\{(x_2, y_2), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_1 = (y_1 - \hat{y}_1)^2$

- Remove (x_2, y_2) for testing.
- Train the model on $n-1$ points: $\{(x_1, y_1), (x_3, y_3), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_2 = (y_2 - \hat{y}_2)^2$

- Rinse and repeat

Return the score:

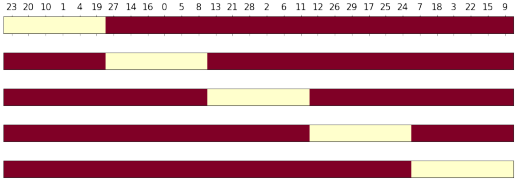
$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i$$

Definition of k-fold CV

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell}\sum_{(x_j,y_j)\in U_i}(y_j - \hat{y}_j)^2$

- Rinse and repeat



Return

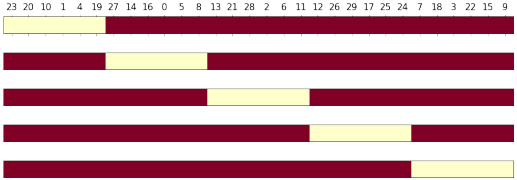
$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k}\mathrm{MSE}_i$$

Setup: k-fold

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{Err}_i = \frac{1}{\ell} \sum_{(x_j, y_j) \in U_i} \mathrm{I}(y_j \neq \hat{y}_j)$

- Rinse and repeat



Return

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{Err}_i$$

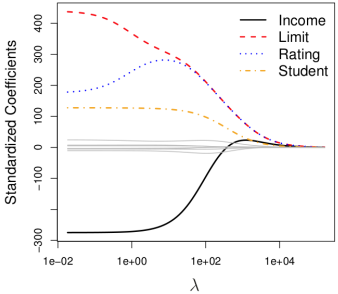
Ridge regression

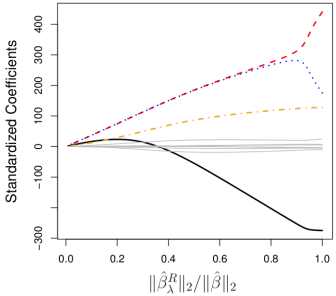
Before:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

After:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$





The lasso

Least Squares:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

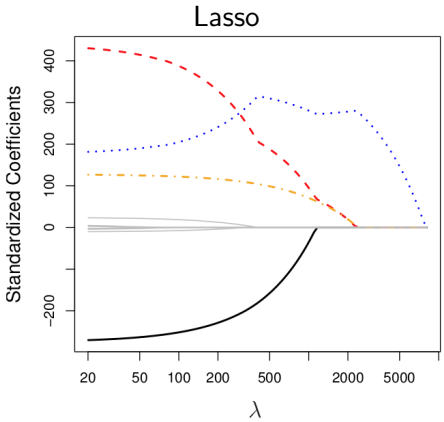
Ridge:

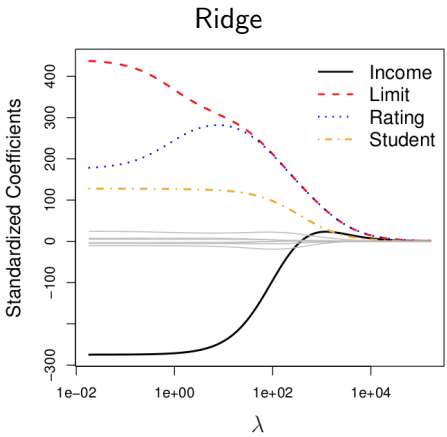
$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$

The Lasso:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\sum_{j=1}^{p}|\beta_j|$$

An example on Credit data set



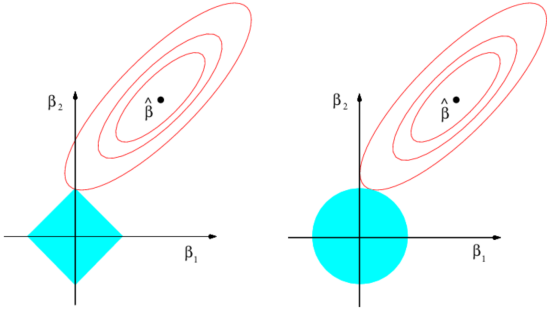


Why the hell lasso can select variable (while ridge cannot)?

Alternative formulation of lasso & ridge regression (play more with ℓ_p)

$$\min_{\beta} \sum (y_i - \hat{y}_i)^2 \text{ where } \sum |\beta_j| \leq s$$

$$\min_{\beta} \sum (y_i - \hat{y}_i)^2 \text{ where } \sum |\beta_j|^2 \leq s$$



Subset selection example: best subset

We train a model using four variables, X_1, X_2, X_3. We're interested in getting a subset of the variables to use. The following table shows the training and test error for the model learned using each possible subset of variables.

	Training MSE	K-fold CV test MSE
null model	0.079	0.614
X1	0.984	0.954
X2	0.172	0.642
X3	0.410	0.533
X1,X2	0.800	0.810
X2,X3	0.696	0.577
X1,X3	0.888	0.779
X1,X2,X3	0.133	0.536

Subset selection example: forward selection

We train a model using four variables, X_1, X_2, X_3. We're interested in getting a subset of the variables to use. The following table shows the training and test error for the model learned using each possible subset of variables.

	Training MSE	K-fold CV test MSE
null model	0.079	0.614
X1	0.984	0.954
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X1,X2	0.800	0.810
X2,X3	0.696	0.577
X1,X3	0.888	0.779
X1,X2,X3	0.133	0.536

Subset selection example: backward selection

We train a model using four variables, X_1, X_2, X_3. We're interested in getting a subset of the variables to use. The following table shows the training and test error for the model learned using each possible subset of variables.

	Training MSE	K-fold CV test MSE
null model	0.079	0.614
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X1,X3	0.888	0.779
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